

1 Agentic Graph Reasoning: Claim-Level Structural 2 Verification for Multi-Hop Knowledge Graph Question 3 Answering

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5 **Abstract**

6 *Keywords:* knowledge graph question answering, large language models,
7 agentic retrieval, ablation study, hallucination, benchmark quality

8 **1. Introduction**

9 Large language models answer factual questions fluently, and often
10 enough they answer them wrongly. The errors are systematic rather than
11 random, and they are hard to catch from the output alone, because a
12 fabricated answer reads in the same register as a correct one; Huang et
13 al. [1] treat this as an intrinsic consequence of how the models are built
14 rather than an incidental defect to be patched away. The problem sharpens
15 once an answer requires composing several facts rather than recalling one.
16 Errors compound, since a wrong resolution at the first hop rarely announces
17 itself downstream, and a question-level correctness check never sees the
18 intermediate steps at all — which leaves a system that answers correctly
19 by accident indistinguishable from one that answers correctly by sound
20 reasoning.

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21 Retrieval-augmented generation [2] addresses the first half of this by
22 grounding the model in retrieved text. It does not address the second.
23 Retrieval runs once, before reasoning begins, so a multi-hop question must be
24 answered from whatever that single query returned, and the passage holding
25 the *intermediate* entity need not resemble the original question — which is
26 the only signal the retriever has to rank on. Representing the corpus as a
27 knowledge graph [3] makes composition a traversal rather than an inference
28 over prose, and GraphRAG [4] exploits this, but it still retrieves statically:
29 a neighbourhood fetched in one shot cannot know what the next reasoning
30 step will need, and is either too small to contain the answer or so large that
31 the relevant edges are buried.

32 Agentic systems close that gap by interleaving the two: the model takes
33 one step, observes what it found, and decides where to look next [5, 6, 7].
34 Retrieval becomes navigation, and the system becomes an agent operating
35 on the graph rather than a pipeline reading from it. Section 2 traces that
36 line of work in detail.

37 One property survives this progression unchanged. Every system named
38 above emits whatever its final generation call produces from the traversed
39 context, and nothing checks that what the answer *asserts* is supported by
40 what the agent actually retrieved. The gap is not fabrication — our measure-
41 ments confirm that graph-navigating systems assert entities that exist in the
42 graph and are reachable along the paths they walked (Section 5.4). It is that
43 an entity can sit in the retrieved subgraph, have been retrieved correctly,
44 and still be the wrong answer: an intermediate node from the reasoning
45 chain, a container where the question asked for a member, or one of several

46 plausible neighbours chosen arbitrarily. A system that verifies retrieval but
47 never verifies assertion cannot catch either case.

48 This paper presents **Agentic Graph Reasoning** (AGR), a knowledge-
49 graph question-answering framework built as an explicit state machine over a
50 constrained, deterministic graph tool API, in which the program orchestrates
51 the loop and the language model is consulted only at bounded decision points.
52 Its distinguishing component is a *Structural Verification Layer*: before an
53 answer is emitted, the draft is decomposed into atomic claims and each is
54 checked against the triples the agent actually traversed. Claims that cannot
55 be grounded trigger targeted re-exploration or are removed, and every answer
56 AGR returns is paired with the supporting triples that justify it.

57 We are precise about what that layer buys, because the ablation of
58 Section 6 is precise about what it does not. Claim verification does not
59 measurably change how often AGR is right; it changes how often AGR is
60 right *for a reason it can exhibit*. The contribution is the output contract
61 — an answer together with the traversed evidence for it — rather than an
62 accuracy gain, and Section 6 reports the null that establishes this at the same
63 length as the effects it does detect.

64 On 400 certified questions from each of WebQSP [8] and ComplexWeb-
65 Questions [9], over a Freebase-derived environment of 2.59 million entities
66 and 8.31 million triples, AGR reaches Hits@1 of 0.755 and 0.522 against
67 four baselines sharing one frozen backbone and one call budget, and it does
68 so at roughly half the language-model calls of the strongest of them. On
69 ComplexWebQuestions it is the only system whose accuracy rises with hop
70 count. Its advantage over the agentic baseline is located rather than asserted:

71 on the questions where the shared cap lets that baseline finish it is slightly
72 ahead, and the entire aggregate margin comes from the 29% and 44% of
73 questions on which the cap truncates it.

74 *Contributions..* This paper contributes:

- 75 1. **AGR**, a state-machine agent over a constrained graph tool API whose
76 answers carry the traversed triples that support them, so the reasoning
77 is auditable rather than merely asserted (Section 3).
- 78 2. A **Structural Verification Layer** that checks atomic claims against
79 traversed adjacency before emission, with its repair routes and their
80 measured firing rates reported as they behave rather than as designed
81 (Section 3.2).
- 82 3. A **controlled comparison** against four baselines on one frozen back-
83 bone under one call budget, so that differences are attributable to
84 architecture rather than to model capacity (Section 5).
- 85 4. A **component-level ablation** with paired significance testing, which
86 finds that removing the planner *improves* accuracy on the shallower
87 benchmark while reducing token use, and that three of four components
88 show no detectable effect (Section 6).
- 89 5. A **hand-read census** of all 259 remaining failures, naming the *echo*
90 *attractor* — a true, grounded entity one hop from the answer that
91 every grounding check passes — and quantifying label defects in both
92 benchmarks (Section 7).

93 Section 2 positions AGR against prior work. Section 3 specifies the frame-
94 work and the verification layer, Section 4 the environment and experimental

95 protocol, and Section 5 the measurements. Section 6 attributes them to
96 components, Section 7 reads what remains, and Section 8 states the limits.

97 **2. Related Work**

98 *2.1. Static Retrieval over Graphs*

99 Retrieval-augmented generation [2] attaches a language model to an ex-
100 ternal corpus by embedding the question, retrieving lexically or semantically
101 similar passages, and letting the model answer from them. Where the an-
102 swer sits in a single retrievable passage this works well; where it must be
103 composed across passages, the ranking signal is measuring the wrong thing,
104 because the passage carrying the intermediate entity resembles neither the
105 question nor anything the system yet knows to look for. Surveys of the
106 graph–language-model intersection identify this as the central limitation of
107 flat retrieval [10].

108 Knowledge graphs make the missing structure explicit, storing facts as
109 typed edges between identified entities so that composition becomes a traver-
110 sal rather than an inference over prose. GraphRAG [4] builds a graph over
111 a corpus and retrieves structured context in place of passages, and related
112 systems extend the idea to long-term memory [11] and temporally-qualified
113 facts [12]. All of them retrieve *statically*: a subgraph is fetched around the
114 linked entities in one shot and handed to the model. Because retrieval runs
115 first and runs once, it cannot know what the second hop will need, and a
116 fixed-radius neighbourhood is either too small to contain the answer or so
117 large that the relevant edges are lost among thousands of irrelevant ones.

118 2.2. *Agentic Navigation*

119 Interleaving retrieval with reasoning removes that constraint. The model
120 takes one step, observes what it found, and decides where to look next;
121 retrieval becomes navigation and the system becomes an agent operating on
122 the graph rather than a pipeline reading from it. This inherits the tool-use
123 and self-reflection patterns established for language agents generally [13, 14,
124 15].

125 Think-on-Graph [5] established the pattern for knowledge graphs with
126 model-guided beam search over relations and entities. Plan-on-Graph [6]
127 added two mechanisms to it: decomposition into sub-objectives, and back-
128 tracking driven by reflection. Reasoning-on-Graphs [7] takes a different route,
129 generating grounded relation paths as explicit plans and then traversing
130 them. Others add tool abstractions [16], cope with incomplete graphs [17],
131 or generalise across structured sources [18], and recent surveys treat the
132 convergence as a research programme in its own right [19, 20].

133 KBQA-o1 [21] is the closest system to AGR’s search strategy and the
134 most important to distinguish from it. It performs agentic knowledge base
135 question answering with full Monte Carlo Tree Search — a policy model
136 proposing actions, a reward model scoring them, and rollouts whose values
137 propagate back through the tree — around a ReAct-style agent generating
138 logical forms against Freebase, with the exploration additionally producing
139 annotations for incremental fine-tuning. AGR does not do this. Its search
140 is guided best-first with a bounded beam and backtracking, and it carries
141 neither rollouts nor value backpropagation; the two are separable on that
142 basis, and we name the difference rather than blur it.

143 2.3. Verification and Self-Correction

144 A parallel line checks generated content after the fact. Chain-of-
145 Verification [22] has a model draft a response, plan verification questions
146 about it, answer those independently, and regenerate; multi-agent debate [23]
147 achieves a related effect by having several instances critique one another.
148 Both establish the premise this paper’s verification layer rests on — that
149 decomposing a draft and checking its parts reduces factual error — but
150 both check against the model’s own knowledge rather than an external
151 symbolic source, and the limits of models correcting themselves without such
152 a source are documented [24]. Hu et al. [25] demonstrate a self-correcting
153 agentic graph pipeline in a clinical setting, and FActScore [26] decomposes
154 generations into atomic facts for evaluation, which is the same decomposition
155 applied as a metric rather than as a control step.

156 What none of them provides is a claim-level check against the *traversed*
157 triples, performed *before* the answer is emitted, inside a graph-navigating
158 agent. That is the gap Section 3.2 fills.

159 2.4. The Attribution Gap

160 A second gap is methodological. Agentic systems are assemblies — de-
161 composition, scoring, backtracking, self-correction — and the literature re-
162 ports them as wholes. When such a system outperforms a static baseline,
163 the published work rarely establishes which mechanism produced the gain,
164 or whether some contribute nothing. Later work then inherits the entire
165 assembly on faith, along with its cost.

166 The exception is instructive. KBQA-o1’s own ablation removes MCTS
167 while leaving every other component intact and drops GrailQA F1 from 78.5

168 to 48.5 [21] — a thirty-point attribution to a single mechanism, obtained
169 precisely because the authors ablated rather than reporting the assembly
170 whole. Section 6 applies the same instrument to AGR, and reports the
171 components for which it detects nothing at the same length as the one for
172 which it does.

173 **3. The AGR Framework**

174 *3.1. State Machine and Tool API*

175 *3.2. Structural Verification Layer*

176 *3.3. Budgets and Termination*

177 **4. Experimental Setup**

178 *4.1. Knowledge Environment*

179 *4.2. Benchmarks, Baselines, and Backbone*

180 *4.3. Metrics and Protocol*

181 **5. Results**

182 *5.1. Accuracy*

183 *5.2. Cost and the Location of the Margin*

184 *5.3. Accuracy by Hop Depth*

185 *5.4. Groundedness*

186 **6. Component-Level Attribution**

187 *6.1. When Planning Hurts*

188 *6.2. Three Components with No Detected Effect*

189 **7. Error Analysis**

190 *7.1. The Echo Attractor*

191 *7.2. Benchmark Label Defects*

192 **8. Discussion and Limitations**

193 **9. Conclusion**

194 **Declaration of competing interest**

195 The authors declare no competing financial interests or personal relation-
196 ships that could have influenced the work reported here.

197 **Data availability**

198 Code, prompts, per-question run records, and the analysis scripts that
199 produce every figure and table in this paper are available at [https://github](https://github.com/sakif-khan/agent-ic-graph-reasoning)
200 [ub.com/sakif-khan/agent-ic-graph-reasoning](https://github.com/sakif-khan/agent-ic-graph-reasoning) under the MIT licence.
201 The knowledge environment is derived from Freebase [3] and is rebuilt from
202 its public source by the scripts in that repository rather than redistributed;
203 the question sets are the public WebQSP [8] and ComplexWebQuestions [9]
204 releases.

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